MIL-C-51540(EA)
21 March 1989
SUPERSEDING
(See section 6)

MILITARY SPECIFICATION

CONTROL MODULE (FOR THE XM84 FILTER UNIT)

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrical control module functioning as a distribution center for 28 volt (V) direct current (dc) and 208 V, 400 Hertz (Hz), 3-phase electrical circuits.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereon, cited in the solicitation.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-19-6906 - Control Module.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-6906 - Control Module.

(Copies of standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-6906 shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings listed thereon.

3.1.2 Components. All components of the control module shall conform to the specifications and drawings listed on Drawing 5-19-6906.

3.2 Manufacture and assembly. The control module shall be manufactured and assembled in accordance with Drawing 5-19-6906.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.4 Performance. The following circuits shall conform to the schematic representations on Drawings 5-19-6907 and 5-19-6908 and shall perform as specified in 3.4.1 through 3.4.7 when the applicable air pressures, voltages, and circuit sequences are applied to specific junctions on the control module and tested in accordance with 4.4.4.



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3.4.1 Interconnecting circuits. There shall be continuity between the circuit points in table I and an open between the circuit points in table II when tested as specified in 4.4.4.2.

TABLE I. Continuity between points

| Pin/connector or test point | to | Pin/connector |
|---|---|---|
| J7 - D | | J4 - D, J7 - A |
| J7 - H | | J4 - G, J5 - F, J5-B, TP - 5, J4 - F |
| TP - 1 | | J5 - C, J4 - E |
| TP - 3 | | J5 - D |
| TP - 4 | | J5 - E |

TABLE II. Open circuit between points

| Pin/connector | to | Pin/connector |
|---|---|---|
| J7 - A | | J7 - B, J7 - C, J7 - H, J4 - A |
| J7 - B | | J7 - C, J7 - D, J7 - H, J4 - B |
| J7 - C | | J7 - D, J7 - H, J4 - C |
| J7 - D | | J7 - H |
| J4 - A | | J4 - B, J4 - C |
| J4 - B | | J4 - C |

3.4.2 System power circuit breaker. There shall be an open circuit between test point 2 and pin E of J4 with the system power circuit breaker open and continuity with the breaker closed, when tested as specified in 4.4.4.3.

3.4.3 PE system circuit breaker. There shall be an open circuit between pins A and G of J5 and test point 2 with the protective entrance (PE) system circuit breaker open and continuity with the breaker closed, when tested as specified in 4.4.4.4.

3.4.4 Power relay circuit. The power relay shall pull in and there shall be continuity between the circuit points in table III when tested as specified in 4.4.4.5.



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TABLE III. Continuity between points (power relay circuit)

| Pin/connector | to | Pin/connector |
|---|---|---|
| J7 - A | | J4 - A |
| J7 - B | | J4 - B |
| J7 - C | | J4 - C |

3.4.5 Indicators. All indicators (see Drawing 5-19-6907 and 5-19-6908) on the module shall light when tested as specified in 4.4.4.6.

3.4.6 Warning system. When tested in accordance with 4.4.4.7, the warning system shall function as follows: The MASK indicator shall flash and the horn shall sound with a pulsating tone, between 0.5 and 1.1 inches of water gauge (iwg). The peak horn level shall be at least 68 dB in at least one octave band between 200 and 5000 Hz. After 15 to 25 seconds, the horn shall silence and the MASK indicator shall stop flashing and remain ON. The MASK indicator shall go out between 0.5 and 1.1 iwg when tested as specified in 4.4.4.7.

3.4.7 DC voltage. The dc voltage shall be 30 $\pm$ 3 volts between test point 2(+) and test point 5(-) when tested as specified in 4.4.4.8.

3.4.8 Loss of power. The MASK indicator shall flash and the horn shall sound with a pulsating tone when the 208 V alternating current (ac) input power is lost when tested as specified in 4.4.4.9.

3.5 Workmanship. The control module shall be free from cracked, bent, or dented metal sections, sharp edges, and foreign matter (dirt, oil, or viscous materials).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not



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relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall apply to repair parts contracts or purchase orders only.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of three packaged control modules.

4.3.2 Inspections to be performed. As determined by the Government, the first article sample may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any control module fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any control module to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the control module assemblies produced by one manufacturer, from the same materials, at the same locations, under essentially the same manufacturing conditions, and at essentially the same time.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling of the lot for examination shall be conducted in accordance with MIL-STD-105 (see 4.4.3.3).

4.4.2.2 For tests.

4.4.2.2.1 Test on unpackaged items. Each control module shall be performance tested (4.4.4).



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4.4.2.2.2 Tests on packaged items. Sampling for unit pack container leakage (4.5.1) and heat seal seam strength (4.5.2) shall be in accordance with MIL-STD-105 using an AQL of 2.5 percent.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample control modules and their packaging shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For tests.

4.4.3.2.1 Unpackaged items. Each control module assembly shall be tested for performance prior to packaging. Any assembly found defective shall be rejected.

4.4.3.2.2 Packaged items. The sample packaging shall be tested in the following order:

(a) Unit pack container leakage (nondestructive)

(b) Heat seal seam strength (destructive)

4.4.3.3 Classification of defects.

(a) Control module (Dwg 5-19-6906).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 1.0 percent defective | |
| | 101 | Component missing |
| | 102 | Improper soldering |
| | 103 | Electrical designations missing or incorrect |
| | 104 | Missing or incorrect voltage warning sign |
| | 105 | Workmanship (3.5) |

(b) Packaging inspection (SPI P5-19-6906).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 4.0 percent defective | |
| | 101 | Item not completely cleaned/dried |
| | 102 | Intimate wrap missing or not as specified |
| | 103 | Spacer pads missing or not as specified |



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| | |
|---|---|
| 104 | Cushioning missing or not as specified |
| 105 | Interior container missing or not as specified |
| 106 | Interior container corners not blunted |
| 107 | Barrier missing or not as specified |
| 108 | Unit container and closure missing or not as specified |
| 109 | Marking missing, incorrect, or illegible |

4.4.4 Performance. The control module shall be performance-tested as follows: The control module chassis shall be connected to power supply negative potential throughout this test.

4.4.4.1 Preparation for test. Position the control module on a work bench so that the various controls can be operated during the test. Open all circuit breakers and turn the POWER switch on the control module to the OFF position. Connect the pressure tap on the control module to an incline manometer (0-3 iwg scale, 0.02 iwg per minor increment) and a vacuum device (aspirator, mechanically actuated bellows, etc) adjustable from 0 to 3 iwg. Adjust the vacuum to 2.0 iwg.

Note: The shielded pressure tap on the control module must be unobstructed.

Test the various circuits of the control module in the exact sequence specified in 4.4.4.2 through 4.4.4.9. Continuity is defined as less than 1-ohm resistance. Open is defined as more than 100,000-ohms resistance.

4.4.4.2 Interconnecting circuits. Use an ohmmeter to test between the circuit points specified. For all but the J7-D to J7-A continuity test (table I), continuity shall be defined as less than 1 ohm resistance. Continuity for the J7-D to J7-A circuit path shall be defined as 2 ohms or less. An open circuit is defined as more than 100,000 ohms resistance.

4.4.4.3 System power circuit breaker. Connect the ohmmeter to test point 2 and pin E of J4. The ohmmeter lead with the positive potential shall be connected to test point 2. Turn the POWER switch on the control module to the ON position. The meter shall indicate greater than 9500 ohms resistance and then continuity (less than 1.5 ohms resistance) as the SYSTEM POWER circuit breaker is depressed. Turn the POWER switch on the control module to the OFF position.

4.4.4.4 PE system circuit breaker. Connect the ohmmeter to pin A of J5 and test point 2, then to pin G of J5 and test point 2. The meter shall indicate an open for both circuits. Depress the PE SYSTEM circuit breaker. The same two circuits shall indicate continuity when tested with the ohmmeter.

4.4.4.5 Power relay circuit. With pin B of J5 at ground potential, apply 28 V dc to pin C of J5, causing the power relay to pull in. Using the ohmmeter, test for continuity at the circuit points specified in table III for conformance to 3.4.4.



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4.4.4.6 Indicators. With Pin B of J5 at ground potential, apply 28 V dc to Pin A of J5. Test the "press to test" feature of all the indicators on the control module by depressing their lens cover. All indicators shall light. Apply ground potential to pin E of J5. The LOW PRESSURE indicator shall light. Apply ground potential to pin D of J5. The OCCUPIED indicator shall light.

4.4.4.7 Warning system. Turn the POWER switch on the control module to the ON position. Slowly decrease the vacuum until the MASK indicator and horn come on. The vacuum reading shall be within the range specified and the warning system shall function as specified in 3.4.6. Measure the sound level of the warning horn in all octave bands from 200 to 5,000 Hz with a sound meter positioned 5 feet from the horn. The sound level shall be as specified in 3.4.6. The horn shall silence and the MASK indicator shall change to steady on within the time specified in 3.4.6. Slowly increase vacuum. The mask indicator shall go out within the range specified in 3.4.6. Continue to increase the vacuum to approximately 2.0 iwg. Depress the lens cover of the MASK indicator. The warning system shall function as specified and cease to operate when the MASK indicator is released. Turn the POWER switch on the control module to the OFF position.

4.4.4.8 DC voltage. Remove all previously applied voltages and signals. Apply a 208 V ac, 3 phase, 400 Hz wye power supply to pins A, B, and C of J7, with pin D of J7 connected to power supply neutral.

CAUTION: Connect pin H of J7 and the control module chassis to the 400 Hz power supply neutral prior to making the above power connection to ensure that the control module chassis remains at a safe electrical potential. Make sure power supply neutral is connected to earth ground.

Connect the positive lead of a suitable voltmeter (such as a voltmeter 0 to 50 v dc $\pm$ 3 percent full-scale accuracy) to test point 2 and the negative lead to test point 5. The indicated voltage shall be as specified in 3.4.7.

4.4.4.9 Loss of power. Turn the POWER switch on the control module to the ON position. Remove the 208 V ac electrical power. The warning system shall function as specified in 3.4.8. Turn the POWER switch on the control module off to silence the warning system and complete the test.

4.5 Packaging tests.

4.5.1 Unit pack container leakage (SPI P5-19-6906). The leakage characteristics of the unit pack container (barrier bag) containing the item shall be determined by subjecting the samples to the hot water technique, test method 5009.2 of FED. TEST METHOD STD. NO. 101.

4.5.2 Heat seal seam test (SPI P5-19-6906). The heat seal seams of the unit pack container (barrier bags) shall be tested in accordance with test method 2024 of FED. TEST METHOD STD. NO. 101, except that the static load shall be 45 +5, -0 ounces, and a 5 percent reduction in this static load shall be permitted when room temperature in test area exceeds 90°F.



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5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P5-19-6906.

6. NOTES

6.1 Intended use. This specification covers a control module used to provide operational status and supply and distribute electrical power to the XM84 filter unit.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required
(c) Any special instructions required when acquired as a repair part (see 4.3)

6.3 Subject term (key word) listing.

Airflow valve
Filter unit

6.4 Supersession data. This specification supersedes Chemical Research, Development and Engineering Center Purchase Description EA-C-1319, dated 7 July 1982.

Custodian:

Army - EA

Preparing activity:

Army - EA

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| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-51540(EA) | CONTROL MODULE (FOR THE XM84 FILTER UNIT) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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